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Effects of Sex Hormones on Cognitive Style in Rats and Men

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This paper presents both a review of empirical studies regarding the interacting effects of sex hormones and socialization on cognitive style in rats and men and a new experimental study of the effects of sex hormones on the white rat. In a study of West African males feminized by a kwashiorkor-induced endocrine dysfunction, it was found that males presenting the gynecomastia symptom had a significantly more feminine cognitive style than control males—the former subjects having lower spatial ability, being more field dependent, with higher verbal and lower numerical ability, and having significantly more feminine scores on two measures of sexual bias than the controls. Evidence is also presented to show that in cases of testicular feminization, Turner's syndrome, and Klinefelter's syndrome—even where the sex of rearing is opposite to the genetic sex—the subjects develop cognitive abilities in line with sex of rearing and socialization. The hypothesis is formed as a result of these studies that, while masculine behavior is probably differentiated in the brain by male hormones neonatally, this process is susceptible to both estrogen feminization and dominant female socialization influences. As there is also considerable evidence to show that normal male rats have higher spatial ability than female rats (who have higher activity levels), it was decided to test this hypothesis by establishing an initial sex difference in spatial learning in white rats and then to attempt to reverse it by administering estrogen and testosterone to male and female rats. A significant sex difference in spatial learning was initially established on the Tolman maze. No significant changes were found in the estrogen-treated males after the first estrogen implant, but a significant change toward the female mean was obtained after the second. There were no significant changes for the testosterone-treated females. It is

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suggested that future studies should make use of male and female rats feminized and masculinized at birth to ensure adequate programming of the brain of opposite sex behavior. It is also pointed out that maze- and geometric-type measures of intelligence for rats and men are not truly representative of the major sex-based abilities, as they require a certain level of spatial ability for solution and hence favor males. It is further observed that spatial ability in males has been biologically adaptive in evolution—which would help to explain the physiological basis of sex hormone differences with respect to masculine and feminine cognitive styles.

INTRODUCTION

The purposes of this paper are to review empirical studies concerned with the interacting effects of sex hormones and socialization on cognitive style in rats and men and to present further experimental evidence concerned with the effects of sex hormones on spatial learning in the white rat. The paper is more specifically concerned with testing the hypothesis that the normal sex differences in cognitive style are largely determined by a two-way relationship between sex hormones and socialization. Evidence will be presented to show that normal male cognitive style (higher spatial, numerical, and field-independence scores) is in part influenced by neonatal androgen programming of the brain and in part by the appropriate male socialization processes. Thus the development of male cognitive abilities depends on successful differentiation of the brain by male hormones, which would be reinforced by masculine socialization processes. Similarly for the female, neonatal estrogen programming of the brain, together with appropriate reinforcement by sex-typed female socialization processes, is held to result in the development of a normal female cognitive style—more field dependent, with higher verbal skills. Evidence will be presented to support the hypothesis that the development of masculine and feminine cognitive style can be reversed—both by opposite sex hormone and socialization influences—while a certain amount of interaction between the effects of sex hormones and socialization occurs.

PREVIOUS STUDIES: SEX HORMONES AND COGNITIVE STYLE

These hypotheses were initially tested in a West African study involving African males with severe kwashiorkor who had been feminized and had developed gynecomastia, a condition which Stuart-Mason (1963) states is due to an inability of the liver to inactivate the normal amount of female hormone, estrogen, found in the male. Thus gynecomastia results from an increase in the male of estrogen which eventually stimulates the growth of

the male mammary gland to resemble that of the female. This condition was noted by Stuart-Mason to have occurred among European prisoners of war held by the Japanese. Trowell (1949) also observed in Africa that gynecomastia and other related clinical abnormalities originated from an acute attack of malnutrition in infancy. Gelfand (1957) noted that gynecomastia is a frequent finding in Africa, while considerable atrophy of the testicles may also occur.

It was therefore considered that estrogen feminization resulting from kwashiorkor-induced gynecomastia would result in the development of a more feminine cognitive style. As noted in the introduction, it is thought that sex differences in cognitive style are influenced by sex hormones and socialization in a two-way relationship. Thus while normal males have a more analytical perceptual style and higher spatial and numerical abilities, normal females tend to have lower abilities in these areas but to have higher verbal skills (MacFarlane-Smith, 1964). Higher spatial ability in the male is especially noted (Swineford, 1948; El Koussy, 1955; Taylor, 1960; MacFarlane-Smith, 1964; Howard and Templeton, 1966; Maccoby, 1967).

It would be expected that males feminized by kwashiorkor would be more vulnerable to the influence of female educational and socialization processes and would thus change psychologically in a feminine direction. As kwashiorkor occurs mainly in infancy after weaning—when the infant child has the highest protein requirements in relation to body weight—estrogen feminization would occur in infancy when normal male cognitive style is most vulnerable to opposite sex hormone influences. As Harris (1964) has stated: "during fetal or neonatal life these hormones seem to act in an 'inductive' way on an undifferentiated brain (as they do on the undifferentiated genital tract), to organize it to a 'male type' or 'female type' of brain," while in adult life "the gonadal hormones act on the central nervous system in an excitatory or inhibitory way, and are thus concerned with the neural regulation of gonadotrophic secretion and with the expression of overt patterns of sexual behavior." Levine and Mullins (1966) noted that "the steroids present in the infant also determine to what hormones the brain centers controlling sex behaviors in the adult will be responsive."

It was therefore hypothesized that these West African gynecomastic males with kwashiorkor-induced endocrine dysfunction would manifest a more global field-dependent perceptual style, in terms of Witkin's theory. Witkin has found that, while normal females have a more field-dependent cognitive style, normal males have a more field-independent cognitive style which also correlates highly with spatial ability (Witkin *et al.*, 1962; Dawson, 1967*a*, *b*). Being more field-independent, normal males are therefore better able to overcome embedding contexts in perception, while females have greater difficulty in both space orientation and visually separating items from

the surrounding field. Witkin has also demonstrated a relationship between harsh socialization, maternal dominance, and the development of a more feminine field-dependent cognitive style in affected sons.

Witkin's predictions concerning the effects of socialization on field dependence were also tested and confirmed in cross-tribal studies in West Africa. The Temne, who have harsher socialization, were found to be more field dependent than the permissive Mende (Dawson 1963, 1967*a*, *b*). Furthermore, sons whose fathers had a large number of wives were found to have a significantly more feminine field dependent cognitive style—due to the decreasing contact with the father with consequent increasing maternal dominance. It was further confirmed that the monogamous family group was more effective than the polygamous family group in ensuring the adequate development of sex-based cognitive style and appropriate identification (Dawson).

Further evidence regarding the effects of socialization on cognitive style comes from Kuckenberg (1963), who showed that normal sex differences—higher verbal abilities in females and higher numerical ability in males (Maccoby)—are related to father absence with a consequent increase in maternal dominance. A matched sample of Harvard undergraduates whose fathers did not go to World War II had higher numerical ability due to increased contact with fathers; those whose fathers went to the war had higher verbal ability due to increased contact with mothers. Thus it was expected that feminized males in the West African kwashiorkor gynecomastia study would also have lower numerical and higher verbal ability.

The above hypotheses were all confirmed, with the West African gynecomastic males having significantly lower spatial ability and a more feminine field-dependent cognitive style than controls. In addition, these subjects had significantly lower numerical and higher verbal ability compared to normal males. They were also significantly more feminine on two tests of sexual bias, while physically they were less muscular than the normal controls (Dawson).

Further evidence for the hypothesized interacting effects of sex hormones and socialization comes from the work of Money (1965) in a study of testicular feminizing syndrome in genetic males (overtly female). Money asks "whether the failure of androgen secretion in late fetal life may somehow heighten the brain's capacity to comply with the demands of female sex assignment and rearing." Thus in these cases feminization of the male is further reinforced by the social assignment of female sex when the body structure is overtly feminine.

In a more recent study, Masica *et al.* (1969) have observed that "testicular feminization (androgen insensitivity) is the clinical analogue of animal rat experiments, in which male off-spring are born with normal-appearing female

genitalia (the internal sexual morphology being male), when cyproterone, a drug having an androgen-antagonistic property, has been administered to the pregnant female at a critical period of gestation." Thus the androgen-insensitive males (who have been exposed to this drug *in utero*) have been clinically feminized in a manner similar to that achieved with male rats by Harris.

Masica *et al.* found that for those genetic male subjects with testicular feminizing syndrome reared as females there was a significantly feminine pattern of scores on the Wechsler (verbal 111.8, performance 102.3), while for subjects reared as males a masculine pattern was obtained (verbal 116.7, space–form 118.0). These results support the hypotheses both in regard to feminization effects of androgen insensitivity and the expected interacting effect on cognitive processes of sex assignment and rearing.

They concluded that "sexual hormonal effects *in utero* may be involved in the development and differentiation of human cognitional patterns; and that, more specifically, perhaps there is a favorable influence of androgenic hormones on the developing brain and potential intelligence," while "the absence of fetal androgens, or androgen insensitivity during fetal life may induce other female–male differences in neural differentiation and organization in addition to those associated with reproductive behavior."

A possible physiological basis of these sex differences in cognitive abilities which would be responsive to the action of androgens and estrogens was also suggested by Broverman *et al.* (1968). Their hypothesis was that cognitive sex differences reflect sex-based differences in physiology which are responsive to gonadal hormones. In particular, they noted that females surpass males on simple overlearned perceptual–motor tasks, while males excel at more complex tasks requiring an inhibition of immediate responses to obvious stimulus attributes (Anastasi and Foley, 1949). They further hypothesized that the sex differences in cognitive abilities are reflections of differences in relationships between adrenergic activating and cholinergic inhibitory neural processes, which in turn are sensitive to the gonad steroid "sex" hormones, androgens and estrogens. They specifically rejected the customary attribution of sex differences "to differences in child rearing." This is at variance with the position taken in the present paper, as it is considered that these sex differences in cognitive abilities not only stem from successful programming of the brain by sex hormones but also are reinforced by socialization, education, and sex-typed child rearing.

Broverman *et al.* argue that because these sex differences in cognitive style are also found in subhuman species such as the rat and the chicken, cultural, "reward" hypotheses are inadequate. However, there is considerable evidence to show that human cultural groups living in extreme desert and snow environments develop adaptive sociocultural systems which encourage

higher spatial skills needed for survival (Dawson, 1963, 1969; Berry, 1966). In addition, Witkin's studies concerning the effects of socialization on individual psychological differentiation are convincing, particularly when they are tested in a cross-cultural context, where harsh-socialization tribal culture results in individual field dependence while individuals from more permissive tribal groups are found to be field independent and to have higher spatial ability (Berry, 1966; Dawson, 1967*a*, *b*). Furthermore, the Kuckenberg study showed remarkable numerical–verbal differences for Harvard undergraduates depending on whether the father was absent. In addition, genetic males with testicular feminization who were also socialized as females obtain somewhat more masculine (i.e., lower verbal and higher space–form) means than do normal females. All of these findings confirm the hypothesis that culture can influence cognitive style and that a two-way interaction is involved.

It is not enough to argue that, because similar sex differences are found in subhuman species, sex differences must be physiological. The human species has developed highly adaptive sociocultural systems which, as Montagu (1968) points out, function as biological adaptations through education and teaching. Thus culturally determined sex-typed behavior and the appropriate cognitive skills which have survival value for different environments are encouraged. While subhuman species must have more physiologically specified controls of behavior to survive, survival at the human level must involve a greater degree of flexibility to cope with environmental problems ranging from snow and desert to tropical jungles. Thus while human cognitive differences must stem in part from the programming of the brain by sex hormones, they also stem from culturally determined socialization and sex-typed behavior systems which vary according to the specific survival needs of different environments.

In addition to testicular feminization, there are two other syndromes involving sex hormone reversals which permit the testing of the hypotheses set out in this paper. These are Turner's and Klinefelter's syndromes. Shaffer (1962), Money (1964), Money and Granoff (1965), Money and Alexander (1966), Money *et al.* (1967), Alexander *et al.* (1964), and Alexander and Money (1965, 1966) have examined the cognitive and physiological effects of gonadal aplasia on Turner's syndrome, which Shaffer states "is characterized by the absence of definite gonads in the presence of female external genitalia," while "secondary sexual characteristics normally produced by estrogen are lacking." Money *et al.* (1967) state that cytogenetically the syndrome is most commonly one of a missing X chromosome (44+X0) or more rarely of mosaicism (e.g., 44+X0/44+XX). While patients with Turner's syndrome have the morphological appearance of females, their gonads are either missing or rudimentary. Evidence has been presented by

Money *et al.* to show that while there is no retardation in total IQ, verbal IQ is significantly higher than performance IQ. This is the same cognitive pattern which occurs with normal females. Further studies have shown extreme space–form difficulties, a lack of direction sense, and a slight disability with numbers. These findings are in line with the expected effects of female socialization, although the lack of secondary female sex characteristics might have led to somewhat less female socialization practices and thus could have resulted in slightly more masculine space–form scores. However, Money (1964) considers that "the most likely interpretation of the divergent specific factors scores in Turner's syndrome is that they represent cognitional deficits that may occur in this cytogenetic condition, as may many other known congenital defects."

Further studies have been carried out by Money (1964) and Money *et al.* (1967) in regard to the cognitive and physiological status of male patients with Klinefelter's syndrome, in which an extra X chromosome is present so that the chromosome count is 44+XXY (the normal male count is 44+XY). Money notes that these patients look like males morphologically, although they may be only weakly virilized and eunuchoid in body build with moderate to severe gynecomastia. Again, in terms of the hypothesis it would be expected that secondary feminization due to a lack of androgens associated with gynecomastia would be related to the development of a more feminine cognitive style, although rearing as a male would be expected to counter this process. The results show some overall IQ deficit, with a mean of 89.5. As the verbal IQ is 89.7 and the performance IQ is 90.3, this is a more masculine pattern than that found in Turner's syndrome patients or in normal females. This would indicate that the female endocrine feminization has not been sufficiently severe to increase the verbal factor, while socialization as a male would have helped to influence the higher space–form factor, or that the Y chromosome plays a role in these abilities.

Thus for both Turner's and Klinefelter's syndromes the Wechsler cognitive pattern is in line with sex of rearing; that is, female and male, respectively, and not genetic sex for the Turner's syndrome cases. These results support the hypothesis in terms of the reinforcement of sex-typed socialization processes, despite the presence in each case of opposite secondary sexual characteristics.

In summary, there is some evidence to support the hypothesis that, for human subjects, the normal sex difference in cognitive style is influenced by sex hormones in infancy, in a two-way relationship with socialization. A two-way relationship implies that estrogen feminization of the male would (particularly in infancy) result in an increasing susceptibility to feminine influences and the subsequent development of more feminine abilities and interests. This would apply to a greater extent where the father is absent,

in a polygamous family group, or where the mother is otherwise dominant. It would also be postulated that normal males who have lower androgen levels would be more susceptible to father absence, harsh socialization, and maternal dominance, while normal males with high androgen levels would be less susceptible.

ANIMAL EXPERIMENTS: SEX HORMONES AND COGNITIVE STYLE

There is considerable evidence from animal studies to show that similar sex differences in cognitive style are found in rats as well as men. A number of studies have shown that male rats learn mazes significantly faster than female rats (Hubbert, 1915; McNemar and Stone, 1932; Tryon, 1931). Krechevsky (1933) found that maze-bright rats tend to develop spatial or directional hypotheses while maze-dull rats rely more on visual solutions. Rosenzweig *et al.* (1960) considered that rats achieve spatial hypotheses by ignoring dominant visual cues and adopting the less obvious hypothesis of location in space. However, as there is also considerable evidence to show that female rats condition more rapidly than male rats (as with the human; Spence and Spence, 1966), it is thought that female rats are unable to solve spatial learning maze problems involving position reversals as well as males, because female rats form response dispositions while male rats form cognitive maps (Tolman *et al.*, 1946). These cognitive maps would facilitate the formation of spatial orientation hypotheses in male rats.

As there is considerable evidence from these animal studies to show that the normal sex difference in spatial skills exists in rats as well as men, it was postulated that if this sex difference in rats could be reversed by the administration of sex hormones, this would support the hypothesis that developing male and female cognitive styles are responsive to sex hormones in rats and men. The experiments concerning spatial ability in rats suggest that the difference is not only in terms of orientation in space but also in terms of spatial visualization. Thus spatial ability in rats appears to have the same dimensions as in man, involving Thurstone's (1944) S_2 and S_3 factors of spatial orientation and spatial visualization.

Further evidence for sex differences in spatial learning in rats comes from the work of Barnes *et al.* (1966), who reported that male rats (Holtzman strain) were significantly better than female rats on a spatial visualization task in a water maze. They found a significantly lower level of errors for normal male rats than normal females on a water maze position reversal task. During training experiments for the present study, a significant sex difference was also found between normal male white rats and normal

females on six trials of the Hebb–Williams closed field test of rat intelligence, which consists of maze-type problems involving spatial visualization. Cowley and Griesel (1963), using the same Hebb–Williams closed field test, found that with protein-deprived male white rats there were significantly fewer errors than with protein-deprived females. They also found that normal control males made significantly fewer errors than normal control females. It is of interest that the protein deprivation had a more severe effect on test performance in males than females, which may have been due to estrogen feminization of the male rat in infancy—as occurred with the West African males in the gynecomastia study (Dawson, 1966). Although there is no evidence to show that protein deficiency causes endocrine dysfunction in females, Cowley and Griesel (1966) found that protein-deprived male rats had significantly smaller adrenal glands and smaller testes than control males, while they again had lower scores on the Hebb–Williams test than normal males. DeLost (1960) also reported that nutritional deficiency to the extent of 3 g per day for male rats caused "complete atrophy of the thyroid, slowing down of the spermatogenesis and of the activity of the interstitial gland of the testicle, morphological injuries in certain zones of the surrenal cortex without loss of weight in the surrenal gland."

While it is thought that one of the dominant abilities of male rats is spatial learning, responsive to sex hormones, there is also evidence to show that a female rat attribute is a higher activity level controlled by the estrus cycle. Significantly higher levels of activity in female rats as measured by the activity wheel were reported by Hitchcock (1925), Richter (1933), and Harris (1964). Richter also found that the higher female activity levels were closely related to the estrus cycle, with the peak of activity occurring at the peak of the cycle. Harris succeeded in reversing this sex-based difference by the implantation of ovarian and vaginal tissues in male rats castrated within 8–10 hours after birth, thus ensuring the programming of a feminine behavior pattern in the brain. He found that the feminized males had higher activity levels on the activity wheel and that, at the peak of the estrus cycle, they reached the same high level of activity as reached by normal female rats.

A study similar to that of Harris was carried out by Gray *et al.* (1969), who endeavored to reverse a sex difference in terms of higher level of emotionality in the male rat. They administered testosterone propionate (TP) at 4 days of age to the study groups and oil placebo to the controls. All groups were tested on the open field test at 75 days of age. The results of the experiment gave clear support to the hypothesis that experimental alteration of the central nervous and endocrine systems toward the male norm changes emotional behavior in a male direction: "TP injected females defecated more and ambulated less in the Open Field than animals injected with oil alone."

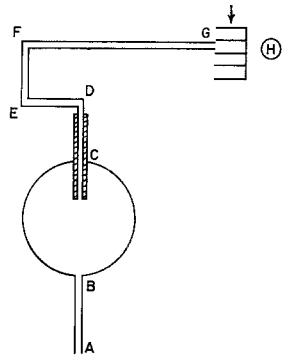


Fig. 1. Apparatus used in preliminary training.

Thus the evidence shows that female rats are better at simple perceptual–motor tasks (such as the operation of the activity wheel), are more easily conditioned, show more spontaneous exploratory activity (Simmel *et al.*, 1965), and are less emotional. On the other hand, the male rat has higher levels of aggression, and inhibition or delay of immediate response tendencies, is less easily conditioned, and has higher spatial learning ability.

PRESENT EXPERIMENT

There is then considerable evidence to show significant sex differences between male and female rats in spatial learning skills. It was decided initially to establish the normal sex difference in spatial learning and then to attempt to reverse it by the administration of the sex hormones, estrogen and testosterone. The Hebb–Williams test was used for preliminary practice to increase test sophistication, while the Tolman sunburst maze was chosen as the measure of spatial learning.

The rats were initially trained to obtain a food reward on the training maze shown in Fig. 1. After these initial trials, the training maze was replaced by the sunburst maze shown in Fig. 2. Rats with high spatial ability were

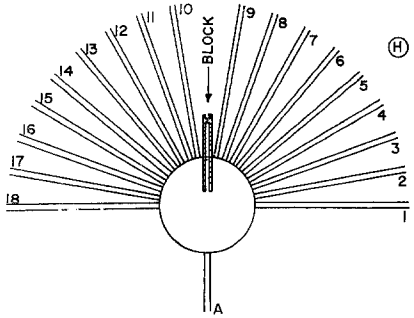


Fig. 2. Apparatus used in the test trial.

expected to choose the paths which pointed toward the position where the food goal box had originally been. Tolman *et al.* obtained significant results with this test, although Gentry *et al.* (1948) failed to replicate the original findings. However, Ritchie (1948) found a strong directional preference for the food side. Gleitman (1955) commented that there may have been difficulty in the Gentry *et al.* experiment due to the use of rats which were experimentally naive. Thus in the present experiment 30 male and 30 female Sheffield white rats were initially tested on 6 successive days after age 100 days in the closed field apparatus described by Hebb–Williams (1946) to ensure that they attained a high degree of sophistication.

The design of the experiment involved next establishing a normal sex difference on the Tolman maze, after which half the male group were given estrogen and half the female group received testosterone. The procedures involved in the Tolman maze test of spatial learning are (except for certain changes noted below) the same as those used by Tolman *et al.* The main hypothesis underlying the study was that masculine cognitive style in rats is differentiated in the brain by male hormones in infancy and hence a higher level of male hormone would be related to higher spatial ability; female hormone would be associated with lower spatial ability. Thus it was expected that the administration of estrogen to male rats would result in a change in the direction of more feminine spatial ability, although it was expected that the estrogen would have less effect with 120-day-old rats than it would have if administered in infancy. The aims of the study were therefore threefold: (1) to investigate the effects of estrogen on male rats, (2) to investigate the effects of testosterone on female rats, and (3) to determine whether any alterations occurred in performance on the spatial task after estrogen and testosterone administration.

The training maze, shown in Fig. 1, consists of a circular unpainted wooden top with elevated paths 2 inches in width. Path AB was 24 inches in length and was used as a starting point. Paths CD, DE, and EF were all 18 inches in length, while path FG was 60 inches long. A stand with sliding food box was located at the end of path FG. When a rat entered a stall, the box was moved in the direction of the arrow. Food pellets for rats were placed in the food stall. After removal, the rats were taken to their cages and fed approximately 30 min later; they were not fed again until after the next trial on the following day. A small battery-powered light was placed at H, 6 inches behind the sliding food box directed along the path FG. Fastened to CD were two pieces of unpainted wood, 18 inches high and 30 inches long, forming an alleyway.

The sunburst maze used in the final test trial is shown in Fig. 2. It consists of the same circular wooden top, with the alley on path CD now blocked and a lamp at point H. The food box and paths DE, EF, and FG

were removed. Twelve 6 ft pine paths were placed around the table top. These are numbered 1–18 in Fig. 2. The light H was beyond path 5 in exactly the same position it was in the training trials. The table top and paths were so arranged that the paths could be rotated after each trial to eliminate tracking. Tolman rotated the table and not the paths, which raised the possibility that the rats would tend to track by going to the same path taken by the previous rat. The room itself was painted white and without cues, while the only illumination was a small battery light at point H.

METHOD

The pretest procedures—the deprivation and feeding schedules—were, except for one or two changes, the same as those used by Tolman *et al.* On day 1 and day 2 the rats were given two trials per day, while on day 3 they received three trials on the training maze. At the conclusion of training on day 3, each rat had run seven times to the food box at G from the starting place at A.

On day 4 the sunburst maze was presented to the rats instead. This was the only trial on day 4. On the test trial each rat starting from point A was allowed to run toward the blocked alley, which they nearly always did, and then to select an alternative path. The first definite path chosen was recorded. This involved an entry to at least 12 inches. The time limit for each rat was 6 min. The criterion Tolman used was that the rat should run out to the end of the path, but this was considered to be too difficult.

The first stage of the experiment, test trial 1, involved contrasting the normal male and female samples to see whether the males showed a significantly higher level of spatial learning than the females. On completion of test trial 1, the groups were subdivided into male study and control groups, and female study and control groups. The male study group received an initial estrogen implant at the age of 120 days. The estrogen pellets (methylstilbestrol, 12 mg) were implanted subcutaneously on the neck dorsally, by means of a pellet implanter. A second estrogen implant was given 14 days after the first implant. The female study group received testosterone pellets (methyltestosterone, 10 mg). As there were a number of deaths in the female study group resulting from the testosterone, the use of this group was discontinued after test trial 2.

Test trial 2 of the experiment was commenced 15 days after the initial estrogen and testosterone implants into the study groups, at the age of 135 days; test trial 3 was carried out 15 days after test trial 2, when the rats were 150 days old, and after the male study group had received their second estrogen implant. The day after the final test trial, two control males and

two estrogenized male rats were randomly selected for purposes of physiological examination.

RESULTS

In general appearance, the estrogen-treated male rats looked more like females, having lost the main external characteristics of male rats, with a considerable reduction in musculature and a decrease in the external size of the testes. The results of the physiological examination of the two control male rats and the two estrogen-treated male rats are set out in Table I.

The data show the body weight of the estrogen-treated males to be much less than that of normal males, while the testes also weigh less. The percent androgen conversion rates indicate that the estrogen-treated males have a much lower percentage of androgen production than the normal control males. This, together with the small size of the testes of the estrogen-treated males, indicates that the process of feminization was successful for these randomly selected rats, at least at the physiological level.

The results for test trials 1, 2, and 3 are set out in Table II. The analysis has been carried out by calculating the immediate entry path means and standard deviations for all samples, using the numbers 1–18 attached to the various sunburst paths as shown in Fig. 2. The exact location of each mean listed in Table II may be checked by reference to the numbering of the 18 paths in Fig. 2. The mean time for entry to the first path over all trials was 53.48 sec, while the mean penetration distance along the first path entered was 2.11 ft.

The results of trial 1 show the normal males to have a mean of 7.27. This is in the expected direction and toward the original goal box, but not directly toward the correct goal path, which is path 5. The normal female mean is, however, 10.41, which is approximately one and one-half paths to the left of the training path, probably indicating a greater degree of female response learning than place learning. Thus the female sample appears to have developed a response disposition in relation to the original alley leading to the food box while the males show a greater degree of spatial or place

Table I. Physiological Data

		Body weight (g)	Testes weight (mg)	Testes size (cm)	Tissue taken (mg)	Percent androgen conversion
1.	CM	254.6	1390, 1389	1.2×0.6	250	31.61
2.	CM	—	1559, 1559	1.2×0.6	251	24.39
3.	OEM	182.8	235, 222.2	0.6×0.6	250	3.51
4.	OEM	181.3	195.5, 195.5	0.6×0.6	250	6.35

Table II. Tolman Maze Results: First Path Entry Criterion

	N	Mean	SD	Analysis of variance *F* ratio	*p*
Trial 1					
Males	26	7.27	3.62		
Females	27	10.41	3.20	3.34[c]	0.01
Trial 2					
Control males[a]	13	7.85	3.82		
Estrogen-treated males[a]	13	9.23	4.10	2.73	ns
Control females	14	11.57	4.32		
Testosterone-treated females	5	11.20	6.94		
Trial 3					
Control males[b]	14	6.29	4.30		
Estrogen-treated males[b]	13	9.54	3.97	5.70	0.01
Control females	14	11.36	3.46		

[a] Difference between means, control males and estrogen-treated males only, not significant ($t = 0.892$).
[b] Difference between means, control males and estrogen-treated males only, significant ($t = 2.295$, $p < 0.05$).
[c] This value refers to t, not to analysis of variance F ratio.

learning—the males have not taken the path next to the training path but have shown a directional preference toward where the food goal box had previously been. The difference between these male and female samples is significant at the 0.01 level, thus confirming the expected normal sex difference in spatial learning.

The results for trial 2, which was carried out 15 days after the initial estrogen implant, show the control male mean to be approximately the same as that for the normal males in trial 1. The estrogen-treated male sample mean, however, shows a change in the expected direction toward that of the female mean. This estrogen-treated male sample change toward the normal female mean is not significant. Similarly, the analysis of variance carried out on the sample means of the control males, estrogen-treated males, and control females is also not significant. The mean for the testosterone-treated female rats is likewise not significantly different from that of the control females.

Trial 3, which was carried out 15 days after the completion of trial 2, using the same samples, shows a slight improvement of the control male sample mean toward the "ideal" path 5. While the estrogen-treated male sample mean shows a significant change from the control male toward the female mean ($p < 0.05$), the control female mean shows no change. The analysis of variance carried out on the three groups in trial 3 is significant.

DISCUSSION

The results of this study have demonstrated the expected normal spatial learning differences between male and female rats, which were previously found by Hubbert (1915), McNemar and Stone (1932), Tryon (1931), Barnes *et al.* (1966), and Cowley and Griesel (1963). While the initial experimental test trial 2 carried out 15 days after the estrogen implant did not show any significant estrogen-treated male sample change, there was a trend apparent which was significant in test trial 3, 30 days later after two estrogen implants. The overall analysis of variance F ratio for test trial 3 was significant at the 0.01 level while the difference between the means of control and estrogen-treated males was also significant ($p < 0.05$). The final trial thus provided some support for the hypothesis that the estrogen-treated male sample would move closer to the feminine controls on the spatial task. The physiological data indicated that the estrogen implants were successful for the randomly selected estrogen-treated males, with a decrease in size of testes and androgen production. Unfortunately, with only five survivors of the testosterone-treated female study group, it was not possible to test adequately the other main hypothesis that masculinization of the female by testosterone would be related to a change toward the male controls. The mean for the remaining five female testosterone-treated study group rats in trial 2 showed no significant change from the female controls.

While there have been some changes in spatial learning for the male estrogen-treated study group toward the female mean, it is considered that these changes would have been much more marked in both test trials 2 and 3 if the experiment had used male rats feminized in infancy. Thus Harris, using castrated male rats which had been implanted with ovarian and vaginal tissues in the first 8–10 hr of life, succeeded in programming feminine behavior in the brain and achieved a feminine estrus activity cycle as measured by the activity wheel. If such a feminized sample had been used in the present study, it would have been expected that the spatial learning mean would have been closer to that of normal females. Similarly, if female rats had had their ovaries removed in the first 24 hr of life along with the administration of testosterone, it would be expected to result in the programming of masculine behavior in infancy, and probably a masculine level of spatial ability in adult life. Using these two study groups with appropriate controls, and studying both spatial learning and activity levels, it would be expected that the castrated males would have lower spatial learning and higher activity levels, while the castrated females would have higher spatial learning and lower activity.

Tolman's Spatial Learning Experiment

The results obtained in these experiments did not support Tolman's

original results, in which part of the sample chose the correct path 5. There was, however, a significant directional preference for males but not for females. It is possible that Tolman's high percentage achieved for path 5 may have been due to the failure to rotate the paths rather than the table, so that path 5 was always in the same relative position, which would probably increase tracking. In the present study, the paths were rotated after each rat. It is proposed in future studies to use simpler measures of spatial learning and spatial visualization skills, as use of the Tolman sunburst maze is time consuming and presents many methodological problems. Thomas and Lyons (1968) have, for example, used a measure of visual field dependency with pigeons, while Rosenblum *et al.* (1965) used a similar technique with monkeys. These techniques could probably be adopted for the study of spatial visualization problems.

The use of the battery light at point H in this experiment should be considered in view of Tolman's contention regarding the formation of cognitive maps in rats and men. Thus he states that "choices can only be made when there are distinctive stimuli in the environment which enable the rat to judge its own location relative to other places in the environment. The light, we believe, performed such a function and was not a mere conditioned stimulus (Tolman *et al.*, 1946)." In a review of his earlier work, Tolman (1966) stated: "The rats had, it would seem, acquired not merely a strip-map to the effect that the original specifically trained-on path led to food, but rather a wider comprehensive map to the effect that food was located in such and such a direction in the room."

Gleitman, in a very interesting experiment involving place learning with rats, found "that without any previous exploration of the room by their own locomotion, most of the rats had formed a cognitive map during their trips in the trolley car." These experiments by Tolman and Gleitman are interesting as they demonstrate the characteristics and complexities of spatial learning in rats. They confirm that rats are capable of forming broad cognitive maps which facilitate place learning. In the present study, however, these effects were limited to male rats, who seemed better able to form cognitive maps than females.

Measuring Sex-Based Cognitive Differences in Rats and Men

There appears to be a need to consider carefully the experimental testing techniques chosen to measure cognitive style in rats as well as men, because of the consistent bias in sex differences found with these tests. There also seems to have been a lack of recognition by many researchers of the existence of these sex differences in cognitive style in both rats and men. This has often led to the selection of both biased samples and to the use of psycho-

logical testing devices which have often produced results not representative of both sexes. Vandenberg (1968) has commented on this problem in regard to male–female factorial differences in abilities. He states that these sex differences would necessitate controlling for sex differences, while many older studies were inadequate because they used samples with unequal numbers of male and female subjects.

The Hebb–Williams closed field test—supposedly a measure of rat intelligence—consists of maze problems which involve a considerable spatial ability component and provides scores that are not equally representative of both sexes. This has been confirmed in the present study. This type of spatial test which so favors the male rat is not an adequate measure of total rat intelligence, as it is only measuring one component of intelligence.

What is needed is a rat equivalent to the Wechsler. Thus while the Hebb–Williams test would represent the spatial component favoring the male, the evidence so far suggests that female rats do better on perceptual–motor activity as measured by the activity wheel, in which there is an established sex difference in favor of the female. Thus if both the Hebb–Williams closed field test and the activity wheel were used together, this might form the beginning of a valid battery which could also be useful for testing the hypotheses outlined in this study.

These results also raise problems of interpretation in regard to Cowley and Griesel's studies in which protein-deprived rats were thought to have significant deficits in intelligence. It seems probable that the Cowley and Griesel results obtained with the Hebb–Williams test did not in fact reflect, as claimed, general changes in intelligence but rather in only one component of intelligence, spatial ability. This interpretation is supported by the sex differences found in their study and also by the fact that, while the normal males showed a significantly higher spatial learning than the normal females, the level for protein-deprived males was lower. As there is no evidence to show that female rats are subject to masculinization resulting from protein deficiency, it is thus possible that this more marked decrease for protein-deprived males may be due to the more severe effects of protein deficiency from birth on the male, resulting in estrogen feminization and lowered spatial skills.

There are also similar problems of biased interpretations with measures of human intelligence, such as the Porteus maze and Progressive Matrices. These involve similar spatial problems, and hence consistent sex differences are found. Podell and Phillips (1959) found that the Porteus maze test loaded consistently on their spatial–decontextualization cluster, while Porteus (1918) reported significantly better performance of males than females on the mazes. Raven (1962) showed a significant difference between matched 13-year-old boys and girls on the Raven Progressive Matrices. In contrasting matched

male and female social science student samples at the University of Hong Kong, we found males to have significantly higher scores on the progressive matrices ($t = 2.290$, $p < 0.05$). MacFarlane-Smith has reported that the progressive matrices load on the general intelligence factor (g), with a low spatial factor (k) for men, but a high spatial (k) loading for women. The reason for this very interesting difference is probably that, since females have lower spatial ability and are more field dependent, they are not as easily able to discriminate geometric forms in solving these test problems, and hence these problems become spatial (k) for females, while being general (g) for males.

Higher Spatial Ability in the Male as Biological Adaptation

We have reviewed earlier some evidence to show that a higher level of spatial ability in the male has been biologically adaptive in extreme subsistence environments in evolution (Dawson, 1963, 1969; Berry, 1966). The ecological adaptation of the nomadic Eskimo and Central Australian Arunta to their respective snow and desert hunting environments has involved the development of adaptive cultural systems emphasizing permissive socialization and individual independence, which, in terms of the theory of Witkin *et al.* (1962), encourages field independence and higher spatial ability. Both the Eskimo and the Arunta have in fact been found to have, as expected, higher spatial ability and field independence. These are adaptive skills which are needed for survival in such extreme biological environments where spatial cues are minimal and some spatial orientation is essential for survival. The Eskimo have also evolved a system of linguistic localizers which have facilitated the solution of spatial problems of orientation in relation to their very open environment. The Eskimo can also draw highly accurate three-dimensional maps in the snow, which must necessitate a very high level of spatial ability (Berry, 1966).

As contrasted with the Temne, who live in a lush jungle environment, both the Eskimo and the Arunta have very advanced art forms—again apparently stemming from these high Eskimo and Arunta spatial skills (Dawson, 1969). The Temne have rigid social organization (paramount chiefs) and harsh socialization, which is adaptive since it seems to produce the high degree of obedience, conformity, and group orientation required for the successful growing and accumulation of rice. As noted in the introduction, this type of harsh socialization results in the development of a lower level of individual spatial ability and more field-dependent abilities. However, these more field-dependent skills are still biologically adaptive for such communities, as higher spatial skills are not required for survival among sedentary agricultural village populations (Dawson, 1967*a*, *b*).

These studies have provided evidence from extreme ecological environments to support the contention that high spatial ability in the male has been biologically adaptive in evolution. In response to specific biological environments, the level of spatial ability required for survival in each environment has been elicited through socialization and related pressures. Thus the Eskimo and Central Australian Arunta, living in snow and desert, have high spatial ability, while the agricultural Temne have a much lower level.

Vandenberg (1968) has also observed that man in evolution gradually improved skills necessary for production by chipping or flaking of stone axes, such as "spatial visualization and mechanical understanding," which, "as well as practice, were required to predict the results of a given blow." Thus man was probably better able to adapt to more diverse ecological environments in evolution because of the development of his high spatial skills, which enabled him to form and utilize primitive tools needed for hunting and the construction of shelters needed in more extreme environments for survival. In addition, man needed good spatial orientation in evolution in order to hunt successfully and to locate the family group in relation to food, water, and territorial restrictions imposed both by natural hazards and by enemies and predators. With the gradual increase of settled agricultural communities over hunting and gathering economics, high spatial ability in the male would not have been so crucial for survival, except among the hunting peoples of the world such as the Eskimo and the Arunta.

In this regard, Masica *et al.* also suggest that "perhaps male–female discrepancies in cognitional achievement in the areas of visual–perceptual organization and spatial relationships are remnants of specialized evolutionary adaptations, reflecting the sharper visual–perceptual skills employed by the male in territoriality and mating."

These suggested biological survival advantages in evolution of higher spatial ability in the human male also help to explain the evolutionary development of a two-way sex hormone relationship with higher male spatial ability and masculine cognitive style, which would have been paralleled by the development of appropriate female perceptual–motor and verbal skills and reproductive activities, also controlled by female hormones in a similar two-way relationship.

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